

6. The DNA double helix structure was discovered by

- (a) James Watson and Francis Crick
- (b) Rosalind Franklin and Maurice Wilkins
- (c) Linus Pauling
- (d) Gregor Mendel

69th B.P.S.C. (Pre) 2023

Ans. (a)

The DNA double helix structure was discovered by James Watson and Francis Crick. James Watson and Francis Crick received Nobel Prize in 1962 for their model of DNA double helix structure.

7. Har Gobind Khorana is credited for the discovery of –

- (a) Synthesis of Protein
- (b) Synthesis of gene
- (c) Synthesis of Nitrogenous
- (d) None of these

39th B.P.S.C. (Pre) 1994

Ans. (a)

The Nobel Prize in Physiology or Medicine in 1968 was awarded jointly to Robert W. Holley, Har Gobind Khorana and Marshall W. Nirenberg for their interpretation of the genetic code and function of DNA in protein synthesis.

8. In laboratory, who did the synthesis of DNA –

- (a) Miller
- (b) Khorana
- (c) D. Weris
- (d) Kelvin

43rd B.P.S.C. (Pre) 1999

Ans. (b)

See the explanation of the above question.

9. Male's gene constituent is –

- (a) XX
- (b) XY
- (c) X
- (d) Y

43rd B.P.S.C. (Pre) 1999

Ans. (b)

The X chromosome is one of the two sex-determining chromosomes (allosomes) in many animal species, including mammals and is found in both males and females. Each person normally has one pair of sex chromosomes in each cell. Females have two X chromosomes (XX) whereas males have one X and one Y chromosome (XY). X of male and X of female give birth to a female child (44+XX) whereas Y of male and X of female give birth to a male child (44+XY). Thus, in human being, the males play a major role in determining the sex of the offspring.

10. Genes on chromosomes are arranged in

- (a) spiral manner
- (b) diffused manner
- (c) linear manner
- (d) More than one of the above
- (e) None of the above

B.P.S.C. School Teacher (Class 9-10) 07-12-2023

Ans. (c)

A gene is a segment of DNA instructions for building a particular RNA or protein molecule and chromosomes are thread-like structure in which DNA is tightly packaged within the nucleus. Genes on chromosomes are arranged in linear manner. The human genome include 21,000 or so genes, spread out along 3 billion base pairs of DNA. The DNA is distributed among 23 chromosomes, of which we have two sets. We inherit one set from each parents.

HUMAN ANATOMY AND PHYSIOLOGY

I. Skeletal and Muscle System

1. Which of the following connects a muscle to a bone?

- (a) Cartilage
- (b) Ligament
- (c) Tendon
- (d) Interstitial fluid
- (e) None of the above / More than one of the above

B.P.S.C. CDPO 2022

Ans. (c)

A tendon is a fibrous connective tissue that attaches muscle to bone. Tendons may also attach muscles to structures such as the eyeball. A tendon serves to move the bone or structure.

2. Stapedius is a–

- (a) bone
- (b) muscle
- (c) nerve
- (d) None of the above

BPSC Agriculture Department-2024

Ans. (b)

The Stapedius is a small muscle in the middle ear that helps to regulate sound.

3. Muscles present in our limbs are

- (a) cardiac muscles
- (b) unstriated muscles
- (c) striated muscles
- (d) None of the above

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Ans. (c)

Muscles present in our limbs are striated muscles. Cardiac Muscles are present in our heart and unstriated muscle found in organs like stomach, blood vessels etc.

4. Which enzyme is not present in muscle?

- (a) Phosphorylase (b) Hexokinase
(c) Glucose-6-phosphatase (d) Glycogen synthase

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Ans. (c)

Glucose-6-phosphatase is not present in muscle. Glucose-6-phosphatase is an enzyme found mainly in the liver and kidneys and plays the important role of providing glucose during starvation.

5. The parts of the skeleton that are not as hard as bones and which can be bent are called as

- (a) vertebrae
(b) carpals
(c) cartilage
(d) More than one of the above
(e) None of the above

68th B.P.S.C. (Pre) 2022

Ans. (c)

The parts of the skeleton that are not as hard as bones and which can be bent are called as cartilage. Cartilage is a tremendously strong and flexible fibrous tissue and it takes many forms and serves multiple purposes throughout the body. There are three types of cartilage–

1. Elastic cartilage
2. Fibre cartilage
3. Hyaline cartilage

6. At the age of 20 years, the number of bones in human body is approximately-

- (a) 100 (b) 200
(c) 300 (d) None of these

41st B.P.S.C. (Pre) 1996

Ans. (d)

The number of bones in an adult human body is 206. The body of a child has about 300 bones at birth, which decreases to 206 bones by adulthood after some bones have fused together. Together these bones form the axial skeleton and the appendicular skeleton.

7. The number of bones in human skull is –

- (a) 28 (b) 30
(c) 32 (d) 40

39th B.P.S.C. (Pre) 1994

Ans. (a)

The human skull (excluding hyoid bone of the neck) consists of 28 bones. It consists of two parts, neurocranium and the facial skeleton (also called the viscerocranium). Neurocranium has 8 bones and there are 14 in the viscerocranium (facial bones) and remaining 6 bones occurs in the middle ear, which is helpful in listening.

8. Leg bone in human body –

- (a) is hollow. (b) is porous.
(c) is solid. (d) is a rivet.

39th B.P.S.C. (Pre) 1994

Ans. (c)

The bone of the foot is solid in the human body.

9. The number of teeth that are developed twice in the lifetime of human being is –

- (a) 4 (b) 12
(c) 20 (d) 28

39th B.P.S.C. (Pre) 1994

Ans. (c)

In human life span, 20 teeth developed twice. Like a human, most of the mammals are diphyodont – It means our 12 permanent molars are monophyodont and others diphyodont. There are four types of teeth– Incisors, canines, premolars and molars.

10. The type of joint found at shoulder is also found at

- (a) elbow (b) hip
(c) ankle (d) knee
(e) None of the above / More than one of the above

B.P.S.C. CDPO 2022

Ans. (b)

The type of Joint found at shoulder is also found in the Hip. The joint present at the shoulder and the hip is known as the ball and socket joint. It allows the movement of the limbs.

11. During expiration the diaphragm becomes

- (a) flattened (b) dome - shaped
(c) normal (d) oblique
(e) None of the above/More than one of the above

B.P.S.C. CDPO 2018

Ans. (b)

Breathing involves two stages, inspiration during which atmospheric air is drawn in and expiration by which the alveolar air is released out. During the expiration the diaphragm relaxes and returns to its domelike shape and the air is forced out of the lungs.